

Submission Summary

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Paper Title

Deep Learning-Based Brain Tumor Detection Using Convolutional Neural Networks on MRI Images

Abstract

One of the most serious and potentially fatal illnesses is brain tumors, for which prompt and precise diagnosis is essential to successful treatment and increased patient survival. Conventional techniques for identifying tumors from magnetic resonance imaging (MRI) scans mainly depend on radiologists' manual analysis, which can be laborious and prone to human error.

Deep learning methods, especially Convolutional Neural Networks (CNNs), have demonstrated considerable promise in recent years for highly accurate automation of medical image analysis. A deep learning-based method for automatically detecting brain tumors from MRI images is presented in this paper. In order to extract pertinent features from MRI scans and categorize them into tumor and non-tumor groups, the suggested system makes use of a CNN architecture.

To enhance performance and generalization, the model is trained on a publicly accessible dataset using suitable preprocessing methods like image resizing, normalization, and data augmentation. According to experimental results, the suggested model outperforms conventional machine learning techniques in identifying brain tumors with high accuracy and efficiency. The system offers a dependable tool to support medical professionals and lessens reliance on manual diagnosis. Future research will focus on expanding the system for multi-class tumor classification and enhancing model performance with sophisticated architectures.

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